When Does Tensorisation Help?

A Pre-Registered Comparison of Hankel-Tensor and Matrix Singular Spectrum Analysis on Synthetic & Financial Time Series

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***Abstract*—A common first step in analysing noisy, multichannel financial data is to split each series into a few interpretable parts, a slow trend, some cycles, and leftover noise. Singular Spectrum Analysis (SSA) does this without assuming a parametric model. Its standard multichannel version stacks the channels into one matrix and decomposes that matrix; a newer family instead stacks them into a three dimensional array (a tensor) and uses tensor decompositions. These tensor methods are often said to recover the underlying parts better, but it is rarely made clear when this should hold, this work gives a precise answer to that. First, a short proof shows that the tensor method is not a larger model: it is the matrix method plus one extra assumption about how channels and time lags combine, so the assumption can help or hurt depending on the data. Then this is tested in two settings: a controlled simulation with known ground truth components, and a real study on stock sector, commodity, and cryptocurrency panels. The simulation shows the prior does not improve component recovery on our main metric; the real study shows a tentative benefit only where a panel's channels are dominated by a single shared factor. The contribution is therefore a tested map of which assumption suits which data, not a claim that one method wins. Future work should fill in the interior of that map with more data regimes.**

***Index Terms*—Singular spectrum analysis, multivariate time series, tensor decomposition, signal separation, financial forecasting.**

I. Introduction

Financial analysis increasingly works with multivariate time series: several related sequences recorded side by side, such as the daily returns of many stocks. These series are noisy, change behaviour over time, and are often effectively short, because the usable stretch between two structural breaks may hold few observations. Models that assume a fixed parametric form and a stable regime tend to do poorly here [1]. One such solution is to first decompose: prior to any forecasting, take the series and divide it up into a few components that are meaningful (trend, cyclical components, and noise) and base the forecasting exercise on those, which is much easier to deal with [2], [3].

Singular Spectrum Analysis (SSA) is the standard tool for this step [2]–[4]. For a given time series, SSA slides a window of length L along it and stacks the windows as the columns of a matrix. That matrix has a special "staircase" (Hankel) structure (every anti diagonal is constant) because neighbouring windows overlap. SSA then takes the Singular Value Decomposition (SVD) of this matrix, keeps the strongest components, groups them into trend, cycles, and noise, and averages each group back into a time series. The appeal is that it is data adaptive and works on short, noisy series, exactly the financial setting [1], [5], [6].

Multivariate SSA (MSSA) extends this to several channels by stacking their Hankel matrices into one tall matrix and again taking the SVD. The idea behind this is simple: with several channels the natural data object is no longer a matrix but a three dimensional array, tensor, indexed by (channel, lag, time). Just as the SVD decomposes a matrix, tensor decompositions decompose this array [7]–[10]. We call the resulting methods Hankel tensor SSA (HSSA). Recent work suggests tensorised embeddings can exploit cross channel structure on short series [11]–[13] and can beat matrix pipelines on noisy biomedical signals [14]. What is missing is a controlled, pre-registered comparison on financial style data that says when the tensor view actually helps.

We commit to four research questions before looking at any results.

- **RQ1.** How exactly is the tensor method related to the matrix method? Is it a strictly smaller model, and what data properties decide how restrictive it is?
- **RQ2.** Under what conditions (such as how much structure the channels share, how noisy the data are, and how short the series are) does the tensor method recover the true components better, and where does any advantage stop?
- **RQ3.** Some tensor decompositions return the individual sources uniquely; others recover only the space the sources span. When the goal is to read off individual sources, does the unique sources decomposition help, and what does that mean for the recovered time series?
- **RQ4.** Do the simulation findings predict what happens on real financial panels with visibly different cross-channel structure?

Our central result is a characterisation, not a recommendation of one method. We prove that tensor SSA is exactly MSSA plus one structural assumption (Section II), then map, on simulated and real data, where that assumption helps and where it hurts. We treat a refuted prediction as just as informative as a confirmed one. Section II sets up the methods and proves the structural result; Section III describes the pre-registered design; Section IV reports both studies; Sections V and VI discuss and conclude.

II. Methods

A. The SSA pipeline

Every method we compare shares the same four step pipeline and differs only in the middle step (Fig. 1). Throughout, $X \in \mathbb{R}^{K \times N}$ is the observed data: K channels, each of length N.

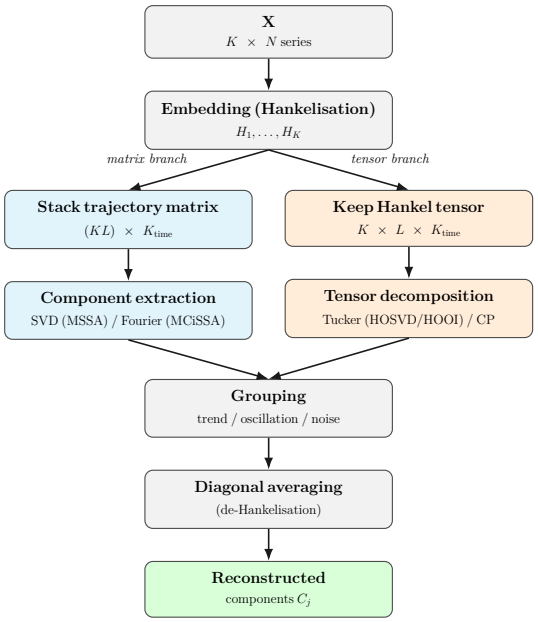


Fig. 1. The shared SSA pipeline. A multichannel series is embedded into one Hankel matrix per channel. MSSA stacks them into one tall matrix and uses the SVD; HSSA keeps the channel as a separate dimension and uses a tensor decomposition (Tucker or CP). Both then group the parts and reconstruct components by diagonal averaging.

1) **Embedding.** Pick a window length L with $1 < L < N$ and set $K_t = N - L + 1$. For channel k, form the Hankel matrix $H_k \in \mathbb{R}^{L \times K_t}$ with $(H_k)_{\ell j} = X_{k,\,\ell+j-1}$; each column is a length-L window.
2) **Decomposition.** Factor the embedded data into a few simple parts. This is the only step that differs between methods.
3) **Grouping.** Assign the elementary parts to interpretable groups ("trend", "cycle", "noise"). Matrix methods group by the spectral content of the singular triples; the circulant method groups by frequency band; tensor methods take one source per factor.
4) **Reconstruction.** Turn each group back into a time series. Because a truncated decomposition is generally not Hankel, we average along anti diagonals (diagonal averaging) to recover each component. All comparisons about the signal are made after this step.

B. The matrix view (MSSA) and the tensor view (HSSA)

Matrix view. MSSA stacks the per-channel Hankel matrices into one block matrix $T = [H_1^\top \ \cdots \ H_K^\top]^\top \in \mathbb{R}^{(KL)\times K_t}$, takes its SVD, and keeps the best rank-r approximation (optimal in Frobenius norm by the Eckart-Young theorem). The model space it can produce is "all matrices of rank at most r". Crucially, channel k and lag ℓ are merged into a single row index of length KL, with no restriction tying them together.

Tensor view. HSSA keeps the channel as its own dimension, forming the tensor $\mathcal{T} \in \mathbb{R}^{K\times L\times K_t}$ with $\mathcal{T}_{k\ell j} = (H_k)_{\ell j}$, and approximates it in one of two ways. The Tucker decomposition [15] (here the orthogonal, energy ordered variant

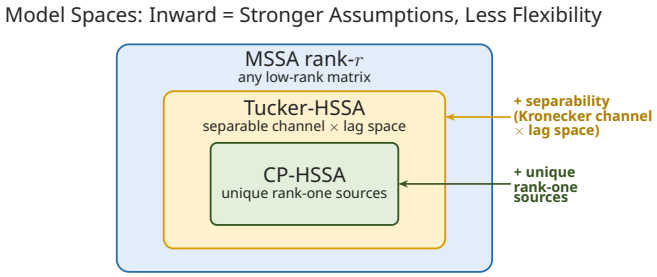


Fig. 2. The three model spaces. MSSA can produce any low-rank matrix; Tucker-HSSA is the subset whose channel lag space factorises (the separability assumption); CP-HSSA is the further subset built from unique rank-one sources. Moving inward trades flexibility for stronger assumptions, which help only when the data satisfy them.

computed by the multilinear SVD (MLSVD), i.e. the higher order SVD (HOSVD) and its iterative refinement [16], [17]) is $\mathcal{T} \approx \mathcal{G} \times_1 U_1 \times_2 U_2 \times_3 U_3$, where each factor U_n has orthonormal columns and (r_1, r_2, r_3) is the multilinear rank; U_1 describes channel patterns, U_2 lag patterns, U_3 temporal patterns. The Canonical Polyadic decomposition (CP, also called Parallel Factor analysis, $\mathcal{T} \approx \sum_{i=1}^{R} \lambda_i\, a_i \circ b_i \circ c_i$, PARAFAC) [18]–[20] is a sum of R rank-one terms ("$\circ$" is the outer product); each term is a candidate source, with channel profile a_i, lag profile b_i, and temporal trajectory c_i.

The structural difference from MSSA is that the channel factor U_1 and the lag factor U_2 are estimated separately. This forces the combined channel lag space to be a Kronecker product $\mathrm{range}(U_1 \otimes U_2)$ rather than an arbitrary subspace of $\mathbb{R}^{KL}$. Informally, the tensor method assumes the way channels mix and the way lags behave can be described independently and then combined; we call this the separability assumption (the separability prior). It is the one extra restriction the tensor view adds.

C. HSSA is a constrained MSSA

Theorem Tucker-HSSA is contained in MSSA.

(a) Flattening the tensor $\mathcal{T}$ along its time dimension gives exactly the MSSA matrix up to a fixed row reordering: $\mathcal{T}_{(3)} = T^\top \Pi$ for a data-independent permutation Π. **(b)** A tensor of multilinear rank (r_1, r_2, r_3) flattens to a matrix of rank at most $\min(r_3,\, r_1 r_2)$. **(c)** Hence with matched budgets ($r_1 r_2 \le r$ and $r_3 \le r$) every tensor the Tucker model can produce is also reachable by rank-r MSSA; containment is strict whenever $r_1 < K$ or $r_2 < L$, since Tucker also requires the channel lag space to factorise as $\mathrm{range}(U_1 \otimes U_2)$.

Sketch. (a) MSSA orders rows by channel-then-lag; the time-mode flattening orders the same entries by lag-then-channel, a permutation Π that leaves rank unchanged. (b) This flattening factors into $U_3\, \mathcal{G}_{(3)}\, (U_1 \otimes U_2)^\top$, meaning its rank is bounded by the lowest rank of its individual matrices. (c) Unlike a generic subspace, a Kronecker-structured subspace is strictly constrained, meaning it only contains vectors that reshape to a $K \times L$ matrix of the restricted rank. A numerical check is in the accompanying code. □

Corollary (Better fit is impossible; better recovery is possible). At matched ranks the Tucker model can never fit the embedded data better than MSSA:
$\min_{M \in \mathcal{M}_{\text{Tucker}}} \|T - M\|_F \geq \min_{\text{rank}(M) \leq r} \|T - M\|_F$.
Here "matched ranks" means the budgets of Theorem II-C(c), so both models keep the same number of components. Two consequences shape how we evaluate the methods. (i) Because Tucker optimises over a smaller set, fitting the observed (noisy) signal can only favour MSSA, so it cannot be evidence for the tensor method. (ii) Yet if the clean signal lies near the Tucker set, projecting onto that smaller set absorbs less noise: a bias variance trade off that can help recovery when the data are short or noisy and hurt it otherwise. One caution: the inequality is stated on the embedded matrix T, but we care about the reconstructed signal after diagonal averaging, and that step is not norm-preserving. A smaller embedded error need not give a smaller signal error (importantly, the reconstructed rank-one components are themselves generally not Hankel) so we report signal level recovery as primary and keep embedded fit only as a diagnostic.

D. Naming sources: CP versus Tucker

The two decompositions differ in what they can identify. CP recovers the individual sources, under a mild condition. To state it we need one notion.

Definition (Kruskal rank)

The Kruskal rank k_A of a matrix A is the largest k such that every set of k columns of A is linearly independent. It is at most the ordinary rank and equals it when the columns are in general position; intuitively, a high Kruskal rank means the columns point in genuinely different directions.

Proposition (Essential uniqueness of CP [21]–[24])

Let A, B, C be the channel, lag, and time factor matrices of a rank-R CP model, with Kruskal ranks k_A, k_B, k_C. If $k_A + k_B + k_C \geq 2R + 2$, the decomposition is essentially unique: any other rank-R CP model of the same tensor has the same rank-one terms, up to relabelling and scaling. In the Hankel embedding the time dimension is large, so the binding part of the condition is the channel mode: the sources must have sufficiently distinct channel profiles. When it holds, CP can name each source by reading off its channel, lag, and time profile. Tucker, by contrast, is rotation-ambiguous (one can rotate the factors and counter-rotate the core without changing the tensor), so it recovers only the subspace the sources span, not the sources themselves. For signal separation this is the meaning of RQ3: if the user wants individual reconstructed sources, CP is the right tool when its uniqueness condition plausibly holds; if only the joint subspace is needed (e.g. for denoising), Tucker suffices.

E. A third comparator: circulant SSA (MCiSSA)

Multivariate Circulant SSA (MCiSSA) [25] stays in the matrix family but fixes the basis in advance. Instead of the data-dependent SVD of T, it (1) estimates the lag-covariance matrix, (2) replaces it by a nearby circulant matrix, whose eigenvectors are always the fixed discrete Fourier vectors, so the elementary components are pure frequencies, (3) groups components by frequency band, and (4) reconstructs each band by diagonal averaging. Because its basis is fixed rather than learned, MCiSSA is neither a special case of MSSA nor of the tensor methods; it should do well when the true sources are well-separated cycles whose frequencies fall near the Fourier grid.

F. What we recover, and how we measure it

We assume a standard linear-mixing model: J latent source series $S \in \mathbb{R}^{J \times N}$ are mixed into the channels by a matrix $A \in \mathbb{R}^{K \times J}$ and observed under noise, $X = A\,S + E$. The true component of source j is the rank-one term $C_j^\star = A_{:,j}\, S_{j,:}$: that source as it appears across all channels. This is what the decomposition should recover, and it is natural in finance: a common "market" factor loaded onto every stock with different weights. How compressible A is decides how well the separability assumption holds. We report two primary metrics, computed after reconstruction: per component Root Mean Square Error (RMSE) against the known component $C_j^\star$, energy normalised; and source correlation, the best match absolute Pearson correlation between a recovered component and its latent source $S_{j,:}$. We keep one diagnostic metric (the reconstruction error of the noisy observation) only to check Corollary II-C; by that corollary it cannot favour the tensor method, so it is never a headline.

III. Experimental Design

We turn the claim into six directional predictions and register them before running anything, so a result going the other way counts as evidence rather than a surprise. Each ties to a research question and to the bias variance mechanism of Corollary II-C.

- **H1** (RQ1). The tensor method never fits the observed signal better than MSSA; any difference is near zero.
- **H2** (RQ2). As channels share more structure (higher ρ, below), the recovery advantage rises.
- **H3** (RQ2). As noise rises (lower signal-to-noise ratio, SNR), the advantage rises, because variance reduction matters more.
- **H4** (RQ2). As the series gets shorter, the advantage rises, for the same reason.
- **H5** (RQ2). When channels are mixed near-orthogonally, so the channel mode cannot be compressed, the advantage disappears.
- **H6** (RQ3). Where CP's uniqueness condition plausibly holds, CP recovers sources better than Tucker.

H2-H4 concern the model classes; the estimators that realise them solve non-convex problems and can land in a poor local optimum, which can blur a predicted trend. We mitigate this with multiple random restarts and many seeds, and flag borderline cases.

A. Synthetic data

The simulation is fully specified so it can be reproduced from the text; Fig. 3 shows one scenario end to end. For

TABLE I
Synthetic factor grid (full cross = 1,200 cells) and the rank/window/seed choices. The seed budget deviates from the registered 50 (deviation D1; see Section V-D).

Factor	Levels
Separability ρ	$\{0, 0.25, 0.5, 0.75, 1\}$
SNR (linear power ratio)	$\{1, 2, 5, 10, 20\}$
Series length N	$\{60, 120, 240, 500\}$
Channels K	$\{2, 3, 4, 6\}$
Mixing geometry	low-rank, moderate, near-orth.
MSSA / CP rank	$r = K$ / $R = K$
Tucker (registered / sensitivity)	$(1, K, K)$ / (K, K, K)
Window length L	$\lfloor N/3 \rfloor$ (synth.), 63 (fin.)
Seeds/cell (run)	50 (ρ=0 and $\rho \geq 0.75$) / 5 (rest)

each scenario we generate K channels of length N. Sources. There are $J = K$ latent sources, each a pure sinusoid $s_j(t) = \sin(2\pi f_j t + \varphi_j)$ of unit amplitude, distinct frequency $f_j = 5+2(j-1)$ Hz, random phase φ_j, sampled at $f_s = 100$ Hz. No amplitude or frequency modulation is used, so the sources are clean cycles. Mixing geometry. The clean data are $X^\star = AS$, with A one of three shapes: low-rank (rank $\lfloor K/2 \rfloor$, strong shared structure), moderate (a generic random matrix), and near-orthogonal (the Q factor of a random QR factorisation, so channels are nearly uncorrelated and the channel mode cannot be compressed). Separability knob ρ. We interpolate toward a rank-one matrix, $A = \rho A_{\text{rank-1}} + (1-\rho)A_{\text{base}}$ (matched in Frobenius norm): at $\rho = 1$ all channels are scaled copies of one profile (the assumption holds exactly); at $\rho = 0$ the channel mode is full. Noise. We add Gaussian noise scaled to a target SNR, defined as a linear power ratio (not in decibels): with σ_{sig} the clean signal's root-mean-square, the noise standard deviation is $\sigma_{\text{noise}} = \sigma_{\text{sig}}/\sqrt{\text{SNR}}$, so SNR$=1$ means equal signal and noise power.

B. Grid, ranks, and inference

We cross five factors (Table I), giving 1,200 scenario cells, and repeat each over many seeds derived deterministically from a master seed. The known number of source signals, K, is utilized as the rank in all cases. The tensor approach is performed at two distinct ranks: the most constrained $(1, K, K)$, which restricts the channel signal mode to a single pattern to test the strongest separability assumption, and the matched (K, K, K), which preserves the tensor structure while allowing the channel mode to utilize K patterns. Comparing the two isolates whether any effect is due to tensorisation or merely to the tightness of the rank.

For each cell we compute, per seed, the paired difference Δ between a tensor method and MSSA (with the sign of each metric chosen so positive means the tensor method is better). This better-is-positive orientation is used only for the directional Wilcoxon and trend tests; the figures and the RMSE values quoted in Section IV-A are raw (un-reoriented) method-MSSA differences, for which lower is better on RMSE (so a positive RMSE difference there means worse than MSSA). We test each cell with the paired Wilcoxon signed-rank test [26], summarise effect size with the rank-biserial correlation and bootstrap 95% confidence intervals, test the ordered predictions (H2-H4) with the Jonckheere-Terpstra trend test [27], [28] (which pools information across cells), and control multiple testing with the Holm step-down correction [29] at family-wise $\alpha = 0.05$. Differences below a pre-set threshold (0.005 diagnostic error, 0.02 correlation, 0.01 energy-normalised RMSE) are treated as practically negligible. Tucker is fit by HOSVD initialisation and higher-order orthogonal iteration (HOOI); CP by alternating least squares (ALS) with several restarts (restarts that disagree also flag a uniqueness violation).

IV. Results

A. Synthetic study

Table II scores each registered prediction; we state the takeaway first.

a) H1 holds (RQ1): No cell shows the tensor method fitting the noisy signal better than MSSA: of 1,200 cells, 343 reject in the predicted (non-positive) direction and 0 in the wrong direction, exactly as Corollary II-C requires. This validates judging the methods on recovery rather than on observed-signal fit.

b) H2-H4 are refuted, and the rank policy is why: At the tight $(1, K, K)$ rank the "higher ρ helps" trend does appear on source correlation (Spearman $r_s = 0.443$, 95% CI $[0.390, 0.488]$), but the tensor method is still worse than MSSA at every ρ: the correlation gap only narrows from -0.35 at $\rho = 0$ to -0.18 at $\rho = 1$. Because the seed top-up (deviation D1) reached the high-ρ cells first, we confirm this on the seed-balanced sub-design where $\rho = 0$ and $\rho = 1$ both carry the full 50 seeds. Across these pairs (each matched on SNR, N, K, and mixing geometry to its ρ=1 counterpart) the gap narrows identically (paired Wilcoxon $p < 10^{-13}$) yet not a single pair shows the tensor beating MSSA at $\rho = 1$. So the trend is genuine but the prior never helps, with no dependence on the seed imbalance. The predicted SNR and length trends (H3, H4) likewise fail as where resolvable they run against the prediction or vanish. Rerunning at the matched (K, K, K) rank closes the tight-rank deficit (per component RMSE within ± 0.03 of MSSA, median near zero) but produces no advantage: the correlation gap stays negative (-0.18 to -0.11) and the right-skewed RMSE mean drifts slightly worse for $\rho \geq 0.25$ (reaching $+0.027$ at $\rho = 1$). The contrast between the two ranks (Fig. 4) is the cleanest finding: the refutation tracks the rank policy, not tensorisation. The tight rank's apparent component RMSE "gain" is an artefact of forcing one channel pattern ($r_1 = 1$), which makes $K - 1$ of the K sources share a single estimate.

c) H5 holds (RQ2): Under near-orthogonal mixing the advantage vanishes as predicted: 154 of 400 cells reject in the predicted direction on source correlation and 0 against (186 vs 2 on component RMSE). When the channel mode genuinely cannot be compressed, no variance reduction offsets the bias.

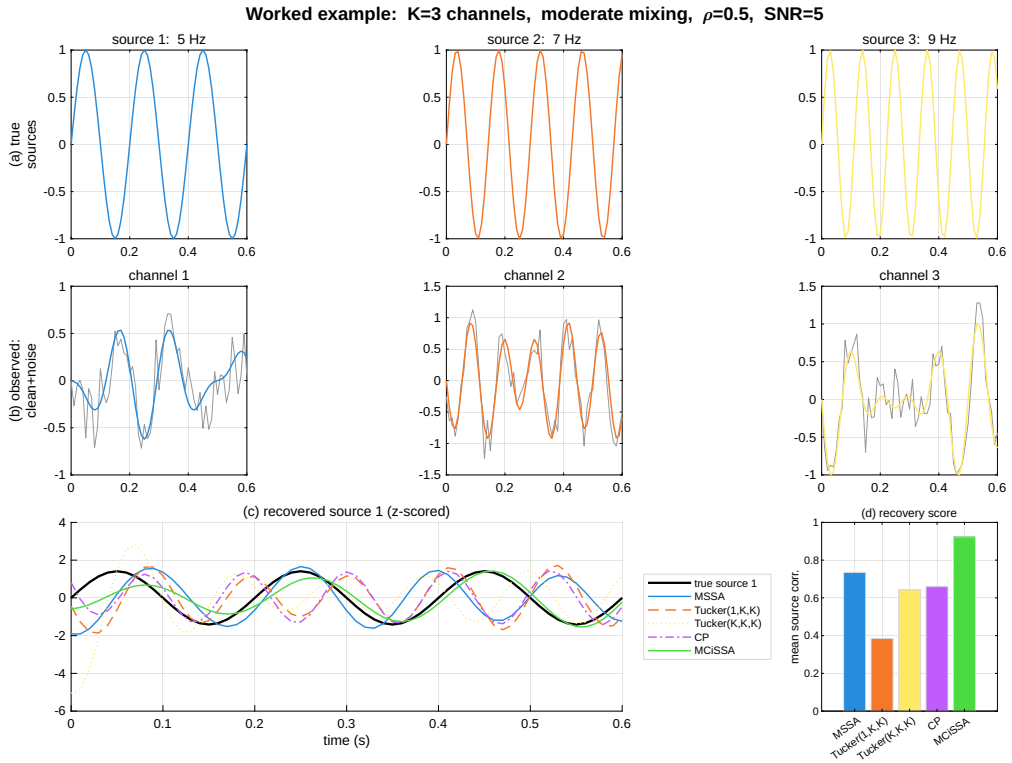


Fig. 3. One worked scenario ($K = 3$ channels, moderate mixing, $\rho = 0.5$, SNR$= 5$). **(a)** the three true sources (clean cycles at $5, 7, 9$ Hz); **(b)** the observed channels (mixed clean signal in colour, plus noise in grey); **(c)** the recovery of source 1 by all five methods, over the truth (black); **(d)** the headline metric, mean source correlation per method. This makes the data-generating process and the metric concrete.

d) H6 holds (RQ3): Where CP's uniqueness condition plausibly holds (high ρ, not near-orthogonal, a region run at the full 50 seeds), CP recovers sources better than Tucker on source correlation in 233 of the 320 proxy cells (median $+0.14$; 74 favour Tucker and 13 are negligible), and the gap is Holm significant in 310 of 320; that 74 cells dissent means the effect is strong but not uniform. This is the signal-level meaning of RQ3: where sources are distinguishable, the unique source decomposition recovers them, and the subspace only one cannot.

e) The circulant comparator: MCiSSA improves source correlation over MSSA at every ρ ($+0.13$ at $\rho = 0$, peaking near $+0.18$, $+0.11$ at $\rho = 1$) while being modestly worse on component-RMSE (median Δ between $+0.03$ and $+0.08$): the fingerprint of a fixed Fourier basis that helps recover cycles but does not improve the reconstruction error.

B. Financial study

Real panels have no ground truth decomposition, so we cannot measure component recovery directly. Instead we ask a downstream proxy: which method's components forecast better out of sample. This changes the construct: better forecasting is at best a downstream consequence of cleaner separation, not a measurement of it (a method can forecast well for reasons unrelated to recovery, and cleaner sources need not forecast better) so we read the financial study as testing whether the map's ordering survives a weaker, indirect proxy, not as a second measurement of component recovery.

We use a walk-forward (expanding-window) protocol and report, per method, the one-step forecast mean-squared error (MSE) and the realised, annualised Sharpe ratio of a simple portfolio built on the forecast [30], with window length $L = 63$ (about a quarter of trading days). We use three baskets chosen to span three structures: **A**, eleven SPDR (Standard & Poor's Depositary Receipts) US sector exchange-traded funds (ETFs); **B**, four macro commodities/currencies (gold, crude oil, USD/CAD, USD/AUD); and **C**, three cryptocurrencies (Bitcoin, Ethereum, Solana). Each price series is converted to log returns (the standard stationary transform); we forward-fill non-trading gaps and trim to a common date range.

The reason a linear-mixing view is reasonable, and the reason we expect different methods to suit different baskets, is structural and decided before seeing the forecasts. Sector ETFs (A) are dominated by a single common "market" factor, so basket A is close to a rank-one channel mode, where the tight separability assumption is plausibly satisfied. Basket B's drivers are distinct (metals, energy, two currency pairs), so its channel mode is hard to compress, the real data analogue of near-orthogonal mixing, where a fixed cyclical basis (MCiSSA) should suit instead. Basket C's three coins share a risk on / risk off factor yet stay distinct enough that CP's uniqueness condition is plausible.

The outcome matches the map, read as alignment rather than a contest (Fig. 5). On basket A the separability assumption is most plausibly satisfied, and the tensor methods give the lowest forecast MSE ($\approx 1.5 \times 10^{-4}$ vs MSSA's 5.4×10^{-4})

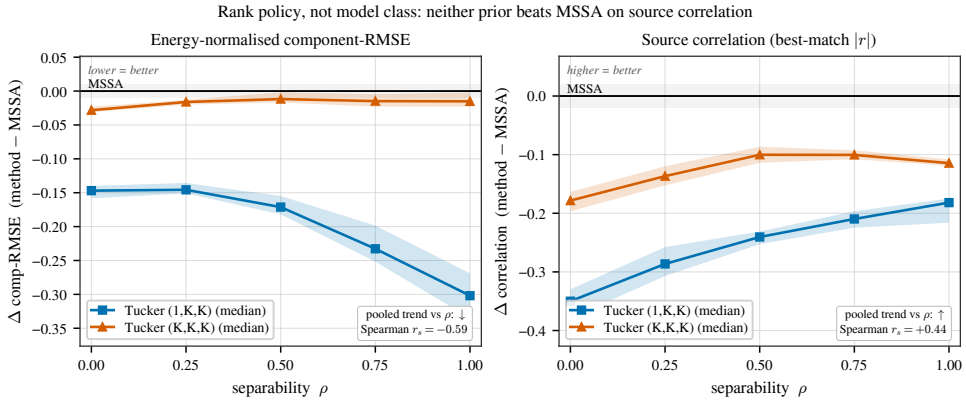


Fig. 4. Rank policy, not model class. Left: energy-normalised component-RMSE difference to MSSA (lower is better); right: source-correlation difference (higher is better); bands are bootstrap 95% CIs. The tight $(1, K, K)$ tensor and the matched (K, K, K) tensor diverge sharply, yet on source correlation neither sits above MSSA at any ρ; the tight rank's lower component-RMSE is interpreted in Section IV-A.

TABLE II

Scorecard for the six pre-registered predictions (synthetic grid). The evidence column avoids quantities distorted by the D1 seed imbalance: it utilizes per-cell median signs and the seed-balanced ρ_0-vs-ρ_1 contrast (evaluating both levels at the full 50 seeds) instead of rejection counts pooled across the imbalanced grid. Furthermore, the pooled Jonckheere-Terpstra trend is assessed solely for direction, as its p-value is inherently seed-weighted.

Prediction	Metric	Verdict	Imbalance-robust evidence
H1	observed-fit not better	supported	median $\Delta \leq 0$ in all cells; 0 wrong-direction rejections
H2	advantage rises with ρ	refuted	trend present (r_s=+0.44) but $\Delta < 0$ at every ρ; seed-balanced ρ_0 vs ρ_1 $p < 10^{-13}$, 0/82 positive
H3	advantage rises as SNR falls	refuted	no positive trend (pooled trend runs the wrong way, $p = 0.005$)
H4	advantage rises as N falls	refuted	no positive trend
H5	advantage vanishes (near-orth.)	supported	$\Delta \leq 0$ at every ρ; 0 wrong-direction rejections
H6	CP > Tucker (identifiable)	supported	CP higher in 233/320 proxy cells, median +0.14

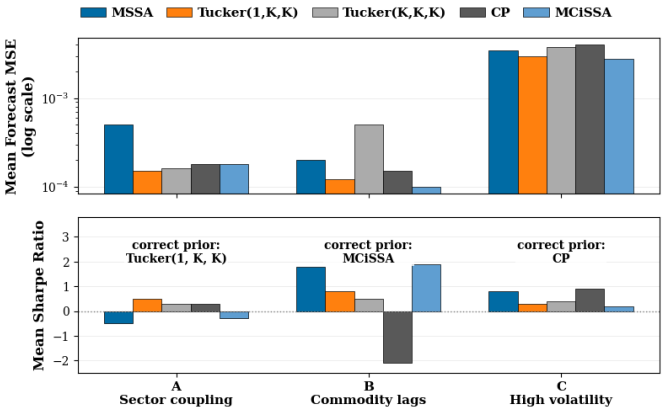


Fig. 5. Walk forward financial benchmark ($L = 63$; basket A 265 folds, B 282, C 199). Top: mean one step forecast MSE (log scale). Bottom: mean annualised Sharpe; the label above each basket names the prior whose assumption best matches that basket's structure, fixed in advance. Of the 24 basket-method-metric contrasts only three survive Holm correction and all are practically negligible, and per fold differences have near zero medians; Section IV-B reads the ordering against the synthetic map.

and the only clearly positive Sharpe ratios (tight tensor +0.49 vs MSSA −0.51). On basket B every Kronecker structure prior is misspecified, the tensor Sharpe ratios fall below MSSA (CP turns sharply negative), and the one method whose assumption fits (MCiSSA's fixed cyclical basis) is the only non-MSSA method above MSSA (+1.91 vs +1.77). On basket C, where CP's uniqueness is plausible, CP attains the highest Sharpe (+0.88 vs +0.77). We treat this as corroboration, not proof: of the 24 contrasts only three survive Holm correction and all fall below the practical threshold, and the headline Sharpe gaps are driven by a few influential folds rather than a systematic per-fold shift (per-fold median $\Delta \approx 0$). A higher Sharpe can in any case reflect a method fitting transient noise in particular folds rather than recovering a more stable underlying signal, which is a further reason we do not read it as a measure of separation quality. With three baskets we place three points on the map; the interior is left to future work.

V. Discussion

A. What differs between the methods

The comparison is clearest as differences in assumptions, not as a ranking. MSSA assumes only low-rank structure in the embedded matrix; it is the safe default and, by Corollary II-C, the best observed-signal fit. Tucker-HSSA adds the separability assumption, which helps only when the channel mode is genuinely compressible and the data are noisy or short enough for the variance saving to outweigh the bias: on our synthetic grid that combination never produces a headline recovery advantage; it appears only on basket A. CP-HSSA adds the stronger assumption of unique rank-one sources; its distinct contribution is not better denoising but the ability to name individual sources, useful when per-source separation is the goal and the sources are distinguishable (basket C). MCiSSA swaps the learned basis for a fixed Fourier one; it

helps recover cyclical sources (basket B) but does not improve reconstruction error.

B. Answers to the research questions

RQ1. That tensor-SSA is a constrained form of MSSA was broadly recognized; the present contribution is to formalize this constraint precisely (Theorem II-C: the $\min(r_3, r_1 r_2)$ bound and the Kronecker condition) and to identify its underlying dependencies: rank choice, separability ρ, and mixing geometry. H1, exhibiting zero wrong-direction rejections, serves as the empirical fingerprint of this behavior. **RQ2.** On the synthetic grid the prior gives no recovery advantage on the headline metric at either rank: the tight rank is strictly worse, the matched rank is indistinguishable to slightly worse. A benefit appears only on real data whose channel mode is empirically near rank one (basket A), and it is absent under near-orthogonal mixing. The clearest practical lesson is that an over-tight rank can erase any benefit. **RQ3.** It is true by construction that CP returns unique rank-one terms while Tucker returns only a subspace; what we add is the signal-level consequence: where sources are distinguishable, CP's components correlate more strongly with the truth (in 233 of 320 proxy cells, median $+0.14$), so CP is the method that lets a practitioner trust an individual recovered source. Where sources are not distinguishable, that uniqueness buys nothing. **RQ4.** The synthetic map does predict the real-data ordering, read as alignment: each basket lands where its structure implies. We regard this as supportive but not conclusive, given three baskets and modest significance.

C. Practical and societal implications

This characterization serves as a practical warning for anyone applying these decompositions to financial data. Our clearest synthetic finding is that an over-tight tensor rank can manufacture an apparent component-RMSE "gain" that is a collapse artefact, not better recovery (Section IV-A). A team that reads such a number as genuine signal separation, and builds a risk model or trading signal on it, inherits a model that looks confident in sample yet rests on a misspecified separability assumption, the kind of hidden fragility that surfaces precisely when the cross channel structure shifts and the assumption breaks. The defensible workflow is therefore to treat the separability prior as a hypothesis to be checked on the data at hand (is the channel mode actually near rank one?) rather than as a free improvement, and to fall back on the safer matrix default when it is not. Read this way, the result is also a small guard against a recurring pattern: promoting a more elaborate method as uniformly better. Because the constrained method yields benefits only within a narrow, identifiable regime, and introduces hidden risks outside of it, any endorsement of its use must be strictly conditional.

D. Relation to prior work and limitations

Our findings sit beside the studies that motivated the work. Joshi et al. [14] report tensor-SSA beating matrix SSA on EEG, a compressible-channel, low SNR setting, the "helps here" side of our boundary; Shi et al. [11] report tensor gains on short series, matching the small N side. The precise statement that tensor SSA equals MSSA plus a separability assumption, with the gain conditional on rank, separability, and mixing, is to our knowledge new; the closest prior remark is that multilinear rank constraints can be more or less restrictive than matrix rank constraints depending on the unfolding [7], [8], [10], [31].

Several limitations qualify the synthetic study. (i) The sources are additive, stationary, finite-rank sinusoids with a single separability knob; real signals violate all of these, which is why the financial study matters. (ii) We fix the window by a rule rather than tuning it. (iii) The seed budget deviates from the registered 50 (deviation D1, our only departure from the pre registration): the high-ρ cells ($\rho \in 0.75, 1$) were run at 50 seeds and the low/mid-ρ cells at 5, except an 82-cell subset of $\rho = 0$ that was topped up to 50 to enable the seed balanced ρ_0 vs ρ_1 contrast. We therefore rest the conclusion on quantities that do not inherit this imbalance: the sign of the per-cell median at each ρ (a median, not a rejection count) and the seed-balanced ρ_0 vs ρ_1 contrast, where both levels carry the full 50 seeds. The pooled trend we read for direction only, since its p value is itself weighted toward the more heavily seeded ρ levels. The imbalance is, however, conservative for our conclusion: the most powered cells are the high-ρ cells where the prior should most help, yet no advantage appears there. (iv) CP's uniqueness condition is applied through a conservative proxy region rather than evaluated cell by cell. (v) The financial study has three baskets covering three structures; the interior of the map is open. (vi) On real panels we treat the mixing matrix A as fixed across each $L = 63$-day embedding window; when the cross-channel factor structure itself shifts inside the window (routine for cryptocurrencies [basket C] and possible in any basket through a regime break) the separability assumption is fitted against a moving target, which can blunt or distort any tensor advantage and is not reproduced by the stationary synthetic generator.

VI. Conclusion

"Does turning multivariate SSA into a tensor method help?" has a precise conditional answer. Structurally, the tensor method is the matrix method plus one separability assumption (Theorem II-C), so a better fit to the observed signal is impossible and any gain must come from better recovery through a bias variance trade-off. Empirically, on a pre-registered 1,200-cell simulation the assumption gives no recovery advantage on the headline metric, and an over-tight rank actively hurts; on three real panels a benefit appears only where a panel's channels are dominated by a single shared factor. Ultimately, the contribution provides a validated map of which assumptions apply to which data environments, explicitly defining the boundaries rather than leaving them asserted. Filling the interior of this theoretical space (through additional mixing structures, non-stationary sources, and tuned window parameters) naturally follows as future work.

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